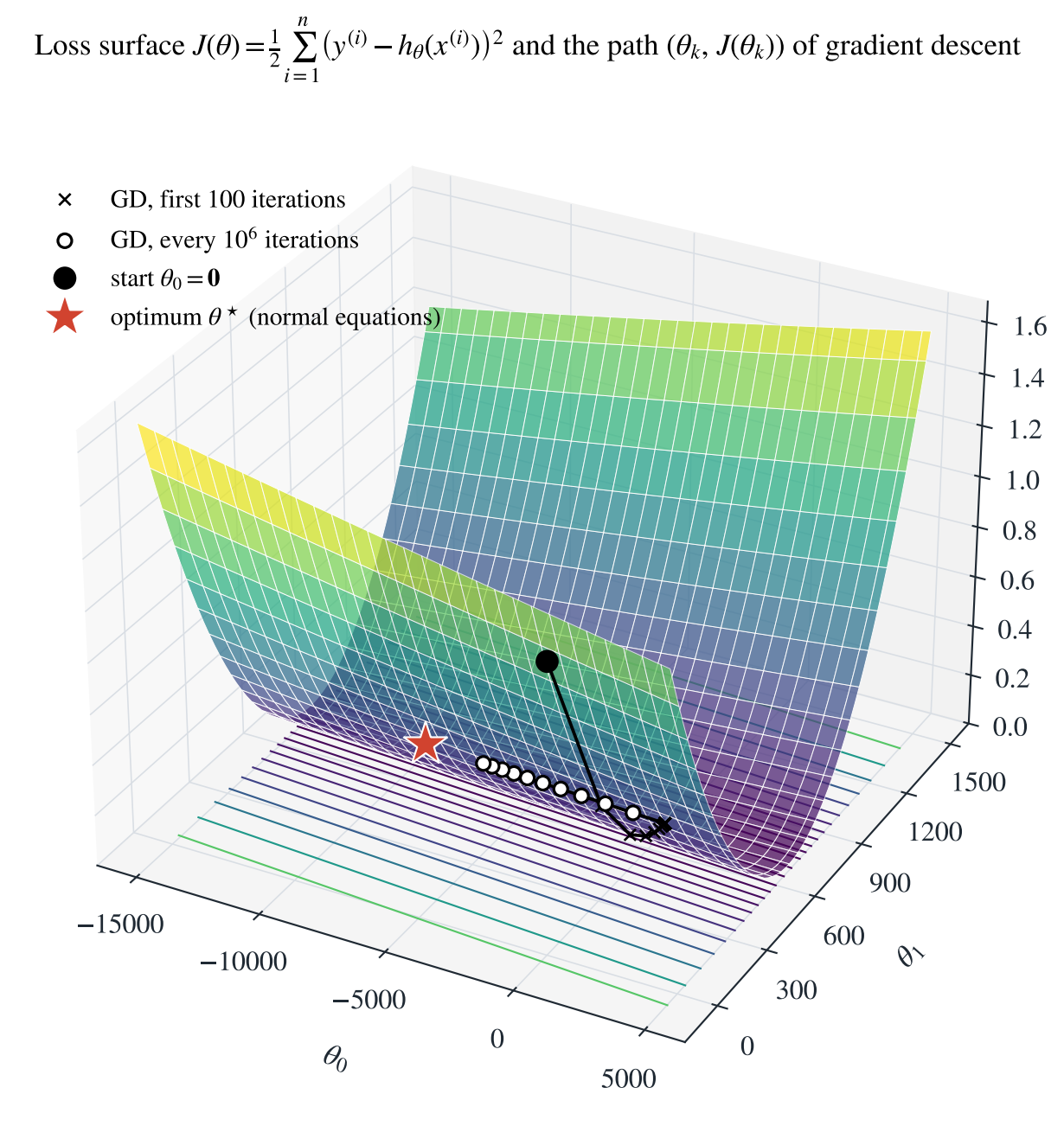

Loss surface $J(\theta) = \frac{1}{2}\sum_{i=1}^{n}(y^{(i)} - h_\theta(x^{(i)}))^2$ and the path $(\theta_k, J(\theta_k))$ of gradient descent
GD, first 100 iterations
GD, every 10^6 iterations
start $\theta_0 = \mathbf{0}$
optimum $\theta^\star$ (normal equations)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1500
1200
900
600
300
0
θ_1
−15000
−10000
−5000
0
5000
θ_0